

Bus Ticketing Application - Problem Statement

Overview

Design and develop a **Bus Ticketing Application** that provides an efficient platform for users to search, book, and manage bus trips, while enabling bus operators to manage their services. Admins will oversee the smooth operation of the system.

Key Components

1. User Role (Passengers)

Core Features:

- **User Registration & Authentication**
 - Register with email/phone and secure password.
 - Login via credentials (Email/Phone + Password).
- **Profile Management**
 - Update personal details.
 - View booking history.
- **Search and Filter Buses**
 - Search for available buses by:
 - Source and destination.
 - Date and time.
 - Bus type (AC/Non-AC, Sleeper/Seater).
 - Filter by price range, ratings, departure/arrival times.
- **Booking Tickets**
 - View seat layout and select specific seats.
 - Real-time seat availability updates.
 - Add passenger details for booking.
 - Apply discount codes (if applicable).
- **Payments**
 - Secure online payments through integrated payment gateways (Credit/Debit card, UPI, Wallets).
 - Payment confirmation and e-ticket generation.

- **Trip Management**
 - View upcoming trips.
 - Cancel booked tickets as per cancellation policy.
 - Receive booking/cancellation confirmations.
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2. Bus Operator Role

Core Features:

- **Operator Registration & Authentication**
 - Registration with company details and verification process.
 - Secure login.
 - **Bus Trip Management**
 - Create and publish new trips with:
 - Source and destination locations.
 - Departure and arrival times.
 - Bus details (type, seat layout, amenities).
 - Pricing and seat availability.
 - Modify trip details (timings, prices, etc.).
 - Cancel trips and notify passengers.
 - **Booking and Seat Management**
 - View real-time seat occupancy.
 - Manage bookings (confirmation, cancellations).
 - **Reports and Analytics**
 - View booking reports.
 - Revenue analysis.
 - Trip performance and customer feedback.
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3. Admin Role

Core Features:

- **User & Operator Management**
 - View, verify, and manage user/operator accounts.
 - Block/unblock accounts for violations.
- **Trip Oversight**
 - Monitor all trips and bookings.

- Modify or cancel trips in emergencies.
 - Enforce platform-wide policies (cancellation, refunds).
 - **Analytics Dashboard**
 - User activity analytics.
 - Operator activity analytics.
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Deliverables

1. Database schema and ER diagrams.
 2. API documentation for the backend
 3. Backend APIs
 4. Unit tests for the backend.
 5. Frontend web application
 6. Panels for user, operator and admin.
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